

E. Coli Culture

This protocol describes the steps required for cultivation of *Escherichia coli* from frozen stock through growth on solid and liquid media. Protocols include isolation of single colonies, expansion in liquid culture, and using optical density as a growth metric. Proper aseptic technique and handling conditions are emphasized to maintain culture viability and prevent contamination. The resulting cultures can then be used for other experimental applications. Refer to [Solid LB Media Preparation](#) to make plates and LB broth. Instructions for making glycerol stocks are also included.

(Day 1: Overnight incubation for solid media)

(Day 2: 2.5–3h for log-phase liquid media, or overnight incubation for stationary-phase liquid media)

Reagents

✓	Name	Quantity	Location/Notes
☐	E. coli frozen stock cultures	Small amount	Store in 30% glycerol at -80°C
☐	LB agar plate	As needed	For streaking on solid media
☐	LB broth	5mL per culture	For liquid media
☐	70% ethanol	Small amount	For sterilizing

☐	50% glycerol - 250 μ L glycerol + 250 μ L ddH ₂ O	500 μ L per glycerol stock	For glycerol stocks
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Equipment

✓	Name	Quantity	Location/Notes
☐	Sterile applicator stick		For transferring colonies
☐	BSC*		Not needed if using ampicillin in LB broth
☐	Glass culture tubes/ Erlenmeyer flasks	As needed	For liquid culture (Day 2)
☐	Spectrophotometer		Monitor growth phase (Day 2)
☐	Shaking incubator		For liquid culture (Day 2-3)
☐	Incubator		(Day 1)
☐	Vortex mixer		Mixing samples

Instructions

Solid Media Colonies (Day 1)

1. In a BSC (if not using ampicillin), remove a small chunk of bacteria from frozen stock using a sterile applicator stick. Streak the bacteria on an LB agar plate
Only briefly remove the stock from the freezer and **DO NOT LET THE BASE STOCK THAW** OR IT WILL KILL THE BACTERIA.
2. Incubate overnight (14-18h) at 37°C.
3. Store at 4°C if not using immediately

Liquid Media Growth (Day 2)

1. In a BSC (if not using ampicillin), transfer 5mL of LB broth into a sterile culture tube
2. Remove one colony from a plate of solid media and inoculate the broth by dipping the colony into the liquid and mixing
3. Close the flask and incubate at 37°C, 300rpm (aeration)

For log-phase growth:

- a. Grow the culture for 2.5–3h, measuring OD600 every 20–30 min to catch mid-log phase (OD600 = 0.4–0.9)
- b. On a growth curve – this is the step rising part of the curve

For stationary phase growth:

- a. Culture can be grown overnight (12–18h)

4. Harvest cells mid-log phase by centrifuging

Glycerol Stock

1. From an overnight liquid culture, combine the following in a 2mL screw top cryotube and mix gently:
 - a. 500 µL liquid culture
 - b. 500 µL 50% glycerol
2. Store at –80C (years) or –20C (months)

Notes here

- Don't need BSC if using ampicillin
- Refer to **Optical Density Monitoring** for measuring OD using the spectrophotometer
- E. coli can grow in a wide range of liquid or solid media but grows faster in rich media like LB (provides amino acids, vitamins, and nucleotide precursors)
- Typically grown in rich media (LB broth or agar plates)
- Growth is usually performed at neutral pH (~7.0)
- OD correlates with the number of viable cells (estimate of culture density)

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- Under these conditions, overnight cultures reach $1-2 \times 10^9$ CFU/mL at saturation
 - At this point (saturation), cells stop dividing due to nutrient depletion and waste buildup
 - Lag phase occurs before exponential growth
 - Log phase is the period of rapid cell division
 - Rich media supports faster bacterial growth
 - Aeration improves growth in liquid cultures

References

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<https://doi.org/10.1016/B978-0-12-818619-0.00070-8>

Explanation of log, lag, and stationary phases for *E. coli*

<https://www.protocols.io/view/preparation-of-bacteria-glycerol-stocks-censtdee.pdf>